distinct vision. This is shown in Figure 7.36. The object distance u should be less than f. The image distance is the near point D. The magnification m of this lens is given by the equation (6.67), $m = \frac{v}{u}$

Substituting, $v = -D$ and $u = -u$, as both the distances are measured to the left of the lens. Hence, $m = \frac{-D}{-u}$

$$m = \frac{D}{u} \qquad (7.59)$$

We can also write the equation for magnification m in terms of focal length f by using lens equation (6.63), $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ in equation (6.67) $m = \frac{v}{u}$ we get $m = 1 - \frac{v}{f}$

Substituting $v = -D$ gives,

$$m = 1 + \frac{D}{f} \qquad (7.60)$$

This is the magnification for near point focusing.

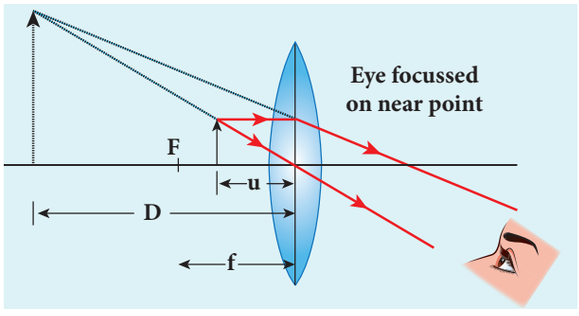
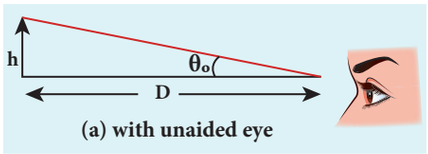


Figure 7.36 Near point focusing

7.6.1.2 Normal focusing

The eye is most relaxed when the image is formed at infinity. The focusing is called normal focusing when the image is formed at infinity. This is shown in Figure 7.37(b). To find the magnification m, if we take the ratio of the height of image to the height of object $\left(m = \frac{h'}{h}\right)$, we will not get a meaningful equation, as the image is of infinite size and it is also formed at infinity. Hence, we can practically use the angular magnification. **The *angular magnification* is defined as the ratio of angle θ_i subtended by the image with aided eye to the angle θ_0 subtended by the object with unaided eye**.

$$m = \frac{\theta_i}{\theta_0} \qquad (7.61)$$



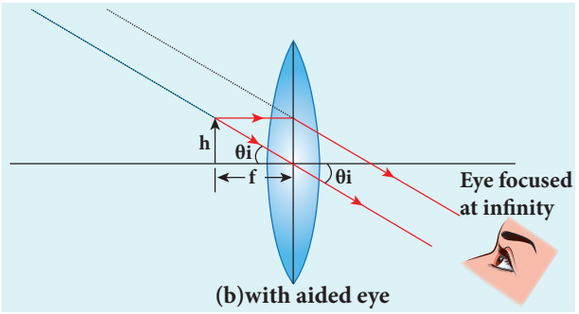


Figure 7.37 Normal focusing

For unaided eye shown in Figure 7.37(a),

$$\tan\theta_0 \approx \theta_0 = \frac{h}{D} \qquad (7.62)$$

For aided eye shown in Figure 7.37(b),

$$\tan\theta_i \approx \theta_i = \frac{h}{f} \qquad (7.63)$$

The angular magnification is,

$$m = \frac{\theta_i}{\theta_0} = \frac{h/f}{h/D}$$

$$m = \frac{D}{f} \qquad (7.64)$$

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